

Ex2-T115952132

Agit Agirman

2026-10-03

```
print(uuid::UUIDgenerate())
```

```
## [1] "e1dd6869-793f-4389-aa85-5f9e5a426d21"
```

Q1

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

```
max(commute); min(commute)
```

```
## [1] 45
```

```
## [1] 18
```

```
table(cut(commute, 5))
```

```
##  
## (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]  
##          5          5          5          5          4
```

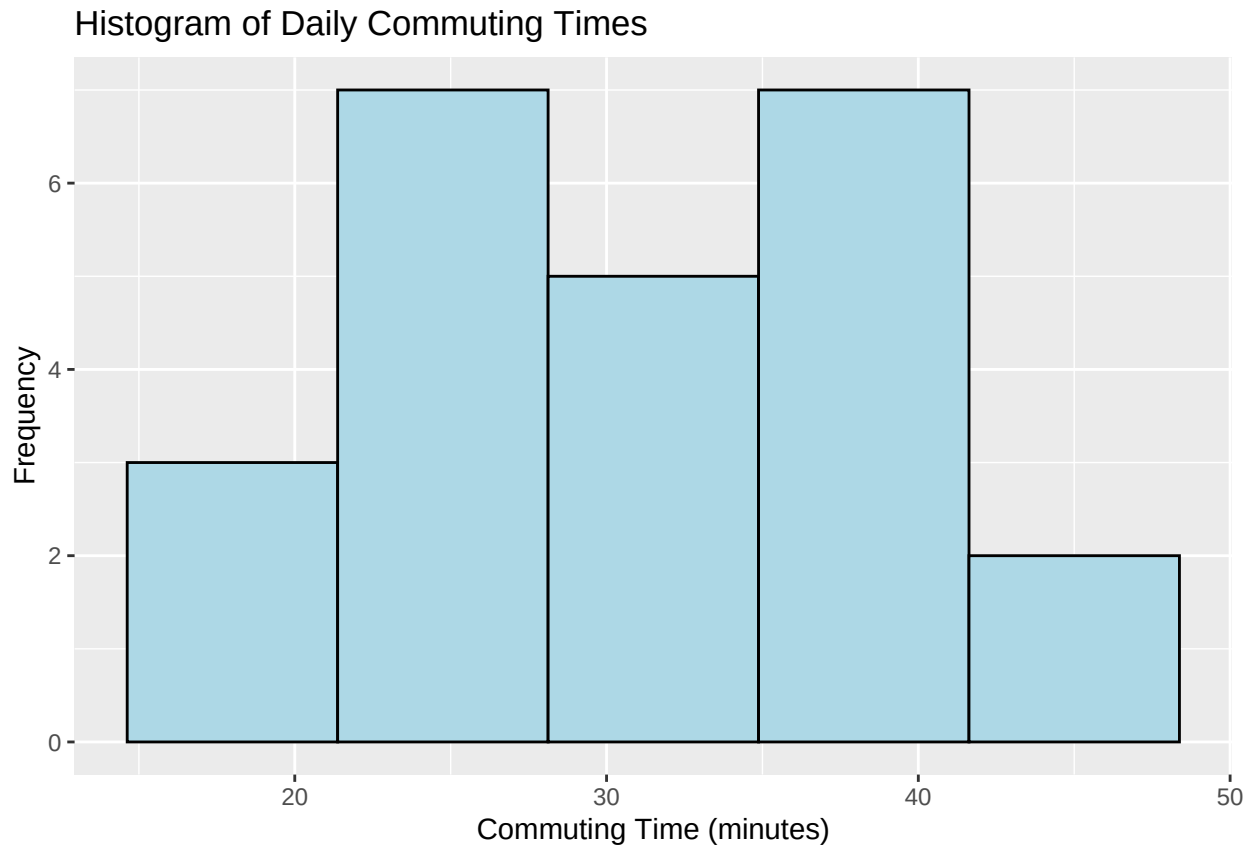
```
cumsum(commute)
```

```
## [1] 18 43 75 116 143 178 200 219 264 302 331 355 386 422 462 488 521 549 570  
## [20] 614 644 681 704 743
```

Q2

```
library(ggplot2)
```

```
ggplot(data.frame(commute), aes(x = commute)) +  
  geom_histogram(bins = 5, color = "black", fill = "lightblue") +  
  labs(  
    title = "Histogram of Daily Commuting Times",  
    x = "Commuting Time (minutes)",  
    y = "Frequency"  
  )
```



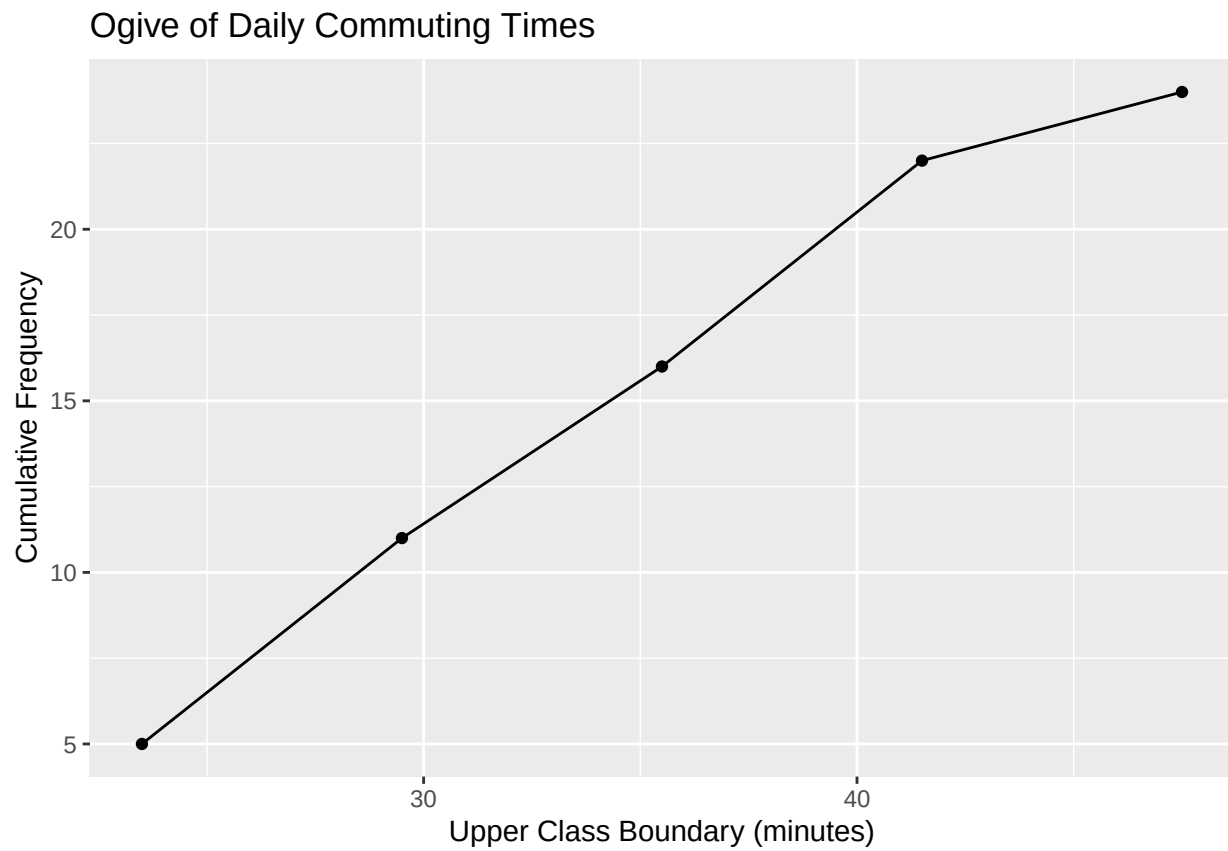
Explanation for Q2: The shape of the distribution is roughly bimodal, with a peak between 20-30 minutes and a second, higher peak between 35-40 minutes.

Q3

```
upper_boundaries <- c(23.5, 29.5, 35.5, 41.5, 47.5)
cumulative_frequency <- c(5, 11, 16, 22, 24)

ogive_data <- data.frame(
  upper_boundaries,
  cumulative_frequency
)

ggplot(ogive_data, aes(x = upper_boundaries, y = cumulative_frequency)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Ogive of Daily Commuting Times",
    x = "Upper Class Boundary (minutes)",
    y = "Cumulative Frequency"
  )
```



Estimation for Q3: Based on the ogive graph, reading the cumulative frequency at the 35.5-minute boundary, approximately 16 students commute less than 35 minutes